

**PRAKTIKUM SISTEM OPERASI**

**MODUL IX  
( SYSCALL )**



**Universitas  
Telkom**

Oleh:

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**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

**DIREKTORAT KAMPUS PURWOKERTO**

**UNIVERSITAS TELKOM**

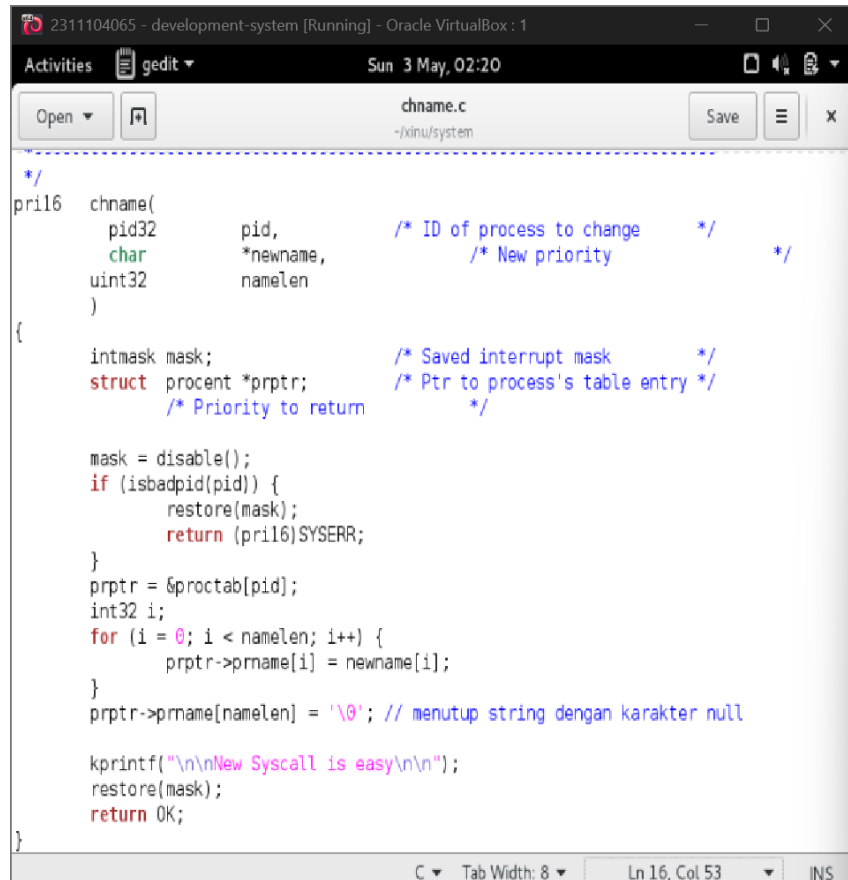
**2026**

## Jurnal

1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5!

Jawab:

- Kode pada xinu/system/chname.c



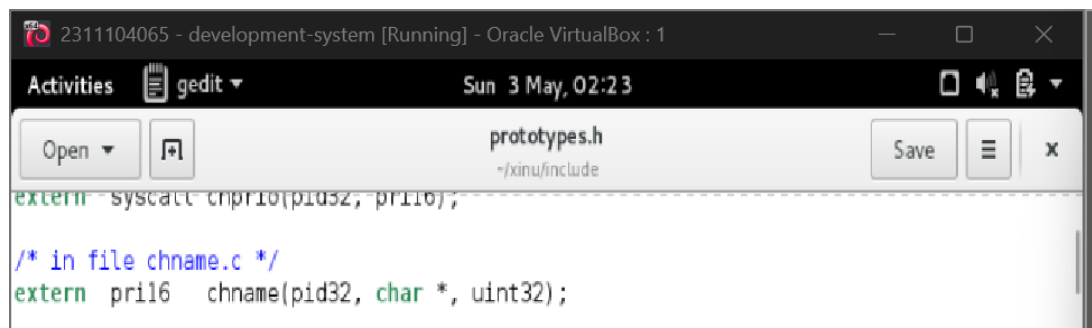
```
2311104065 - development-system [Running] - Oracle VirtualBox : 1
Activities gedit Sun 3 May, 02:20
chname.c
~xinu/system

*/
pri16 chname(
    pid32 pid, /* ID of process to change */
    char *newname, /* New priority */
    uint32 namelen
)
{
    intmask mask; /* Saved interrupt mask */
    struct procent *prptr; /* Ptr to process's table entry */
    /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pri16)SYSEERR;
    }
    prptr = &proctab[pid];
    int32 i;
    for (i = 0; i < namelen; i++) {
        prptr->pname[i] = newname[i];
    }
    prptr->pname[namelen] = '\0'; // menutup string dengan karakter null

    kprintf("\n\nNew Syscall is easy\n\n");
    restore(mask);
    return OK;
}
```

- Kode pada xinu/include/prototypes.h

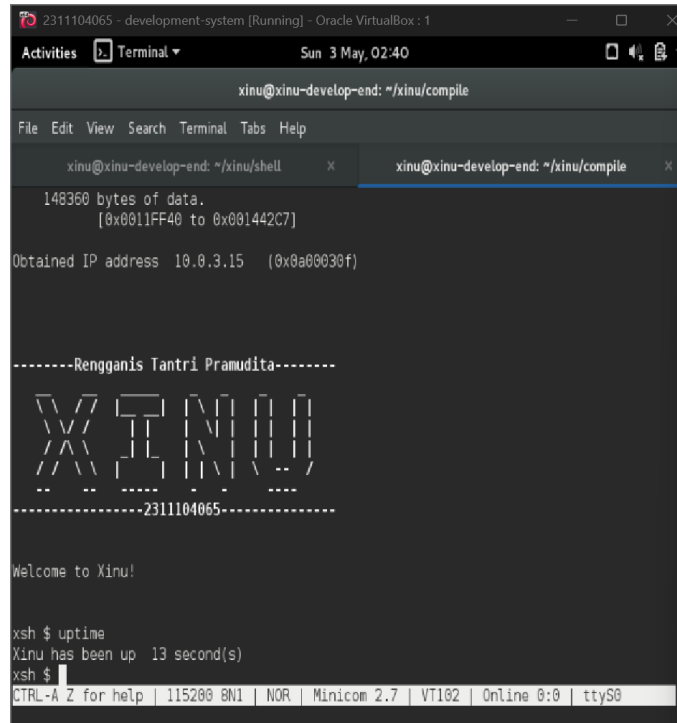


```
2311104065 - development-system [Running] - Oracle VirtualBox : 1
Activities gedit Sun 3 May, 02:23
prototypes.h
~xinu/include

extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern pri16 chname(pid32, char *, uint32);
```

- hasil



```
2311104065 - development-system [Running] - Oracle VirtualBox : 1
Activities Terminal Sun 3 May, 02:40
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/xinu/shell xinu@xinu-develop-end: ~/xinu/compile
148360 bytes of data.
[0x0011FF40 to 0x001442C7]
Obtained IP address 10.0.3.15 (0x0a00030f)

-----Rengganis Tantri Pramudita-----
XINU
-----2311104065-----

Welcome to Xinu!

xsh $ uptime
Xinu has been up 13 second(s)
xsh $
CTRL-A Z for help 115200 BN1 NOR Minicom 2.7 VT102 Online 0:0 ttyS0
```

2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input -

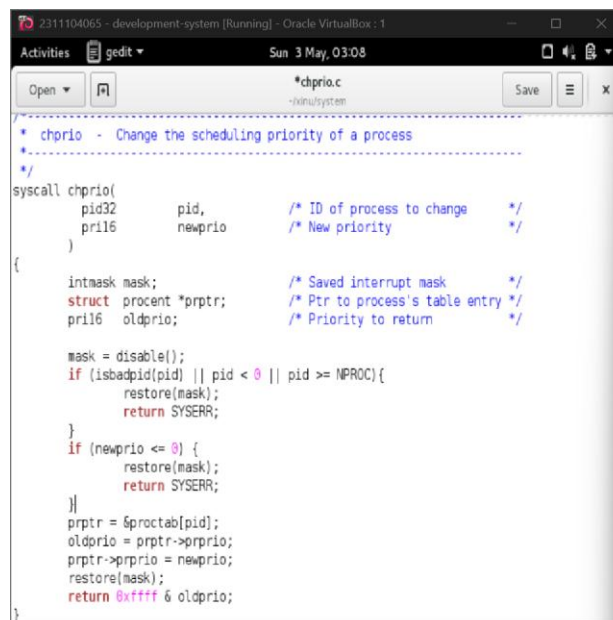
- Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
- Pastikan prioritas adalah bilangan yang positif

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- make clean
- make

Jawab:

- kode pada xinu/system/chprio.c



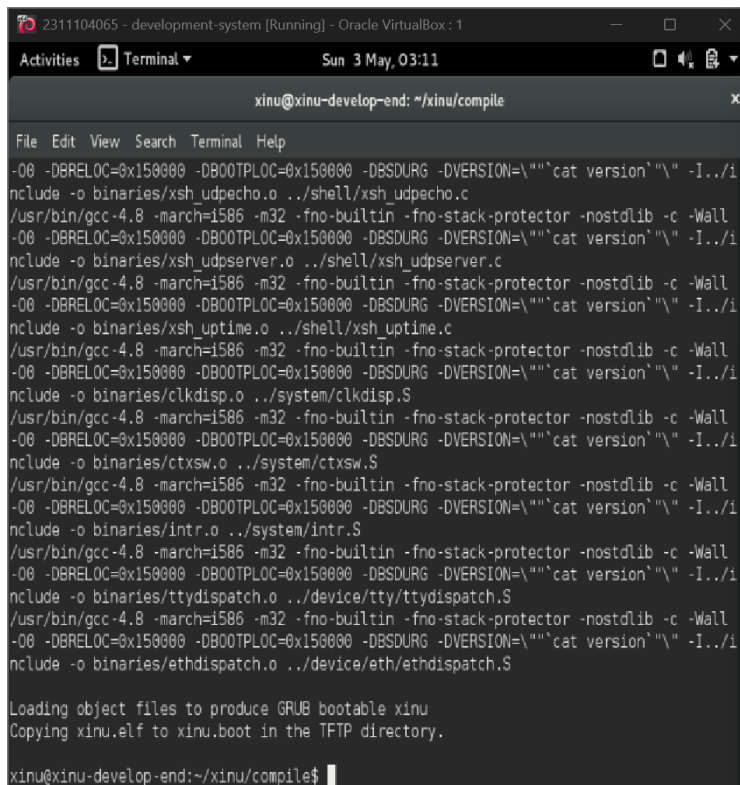
```
* chprio - Change the scheduling priority of a process
*/
syscall chprio(
    pid32    pid,      /* ID of process to change */
    pri16    newprio   /* New priority */
)
{
    intmask mask;      /* Saved interrupt mask */
    struct proctab *prptr; /* Ptr to process's table entry */
    pri16 oldprio;      /* Priority to return */

    mask = disable();
    if (isbadpid(pid) || pid < 0 || pid >= NPROC){
        restore(mask);
        return SYSERR;
    }
    if (newprio <= 0){
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    restore(mask);
    return 0xffff & oldprio;
}
```

- make clean

```
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$
```

- make



```
2311104065 - development-system [Running] - Oracle VirtualBox : 1
Activities Terminal Sun 3 May, 03:11
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_udpsvr.o ../shell/xsh_udpsvr.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/clkdisp.o ../system/clkdisp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.
xinu@xinu-develop-end:~/xinu/compile$
```

### 3. Lakukan hal-hal berikut ini

- Edit xsh\_uptime.c Tambahkan kode berikut

```
/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

// tambahan code
chprio(5,33) // memanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5
```

- Compile source code tersebut dengan perintah
  - make clean

```
xinu@xinu-develop-end:~/xinu/compile$ make clean
removing .o files
removing configuration files ...
make[1]: Entering directory '/home/xinu/xinu/config'
rm -f config lex.yy.c y.tab.c
make[1]: Leaving directory '/home/xinu/xinu/config'
removing xinu ...
xinu@xinu-develop-end:~/xinu/compile$
```

- make

```
2311104065 - development-system [Running] - Oracle VirtualBox: 1
Activities Terminal Sun 3 May, 03:41
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/xinu/shell xinu@xinu-develop-end: ~/xinu/compile
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_udpserver.o ../shell/xsh_udpserver.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/clkdisp.o ../system/clkdisp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"cat version\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.
xinu@xinu-develop-end:~/xinu/compile$ sudo min
```

- Jalankan perintah ps

- xsh \$ ps

- perhatikan prioritas proses dengan id = 5

```
xsh $ ps
```

| Pid | Name         | State | Prio | Ppid | Stack      | Base       | Stack | Ptr | Stack | Size | Msg | Content |
|-----|--------------|-------|------|------|------------|------------|-------|-----|-------|------|-----|---------|
| 0   | pmull        | ready | 0    | 0    | 0x005FDFFC | 0x00FFFED8 | 8192  | 1   | 3     |      |     |         |
| 1   | ipout        | wait  | 500  | 0    | 0x005FBFFC | 0x005FBF30 | 8192  | 0   | 0     |      |     |         |
| 2   | netin        | wait  | 500  | 0    | 0x005F9FFC | 0x005F9ED0 | 8192  | 0   | 0     |      |     |         |
| 4   | Main process | recv  | 20   | 3    | 0x005E7FFC | 0x005E7F34 | 65536 | 0   | 0     |      |     |         |
| 5   | shell        | recv  | 50   | 4    | 0x005F7FFC | 0x005F79AC | 8192  | 0   | 0     |      |     |         |
| 6   | ps           | curr  | 20   | 5    | 0x005F5FFC | 0x005F5FC4 | 8192  | 0   | 0     |      |     |         |

- Jalankan uptime
  - xsh \$ uptime
  - Perhatikan hasil perintah tersebut

```
xsh $ uptime
uptime: 7 seconds
```

- Jalankan ps
  - xsh \$ ps
  - perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFECC      8192 1      3
1 ipout           wait  500   0 0x005FBFFC 0x005FBF30      8192 0      0
2 netin           wait  500   0 0x005F9FFC 0x005F9ED0      8192 0      0
4 Main process    recv  20    3 0x005E7FFC 0x005E7F34     65536 0      0
5 shell           recv  33    4 0x005F7FFC 0x005F79AC      8192 0      7
8 ps              curr  20    5 0x005F5FFC 0x005F5FC4      8192 0      0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0: Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh\_uptime.c

```

        return 1;
    }
    // tambahan kode
    chprio(5,-3);
    secs = clktime; /* Total seconds since boot */

    printf("uptime: %u seconds\n", secs);
    return 0;
}

```

```
xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pmull                 ready  0    0 0x005FDFFC 0x00FFFED8 8192 1 3
1 ipout                 wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin                 wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process          recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell                 recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps                    curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
uptime: 33 seconds

xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pmull                 ready  0    0 0x005FDFFC 0x00FFFED8 8192 1 3
1 ipout                 wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin                 wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process          recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell                 recv  33   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps                    curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:2 | ttyS0
```

- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”

```

        return 1;
    }
    // tambahan kode
    chprio(3000,3);
    secs = clktime; /* Total seconds since boot */

    printf("uptime: %u seconds\n", secs);
    return 0;
}
```

```

xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pnull                 ready  0    0 0x005FDFFC 0x00FFFE08 8192 1 3
1 ipout                  wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin                  wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process           recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell                  recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps                     curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
uptime: 5 seconds

xsh $ ps
Pid Name                State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pnull                 ready  0    0 0x005FDFFC 0x00FFFE08 8192 1 3
1 ipout                  wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin                  wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process           recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell                  recv  50   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps                     curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0

```

Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument  
(dibuktikan dengan menggunakan perintah ps)